Technology: monitoring infrastructure

Types of clouds

- **Public cloud**
- **Private cloud**
- **Hybrid cloud**
- **Rain cloud**
- **Cloud services**
  - SaaS (business user)
  - PaaS (technical development)
  - IaaS (IT technical support)
  - FaaS (microservices, pay per execution, event-driven, auto-scaling, ecosystem).



Virtualized resources are shared publicly.

Supports multiple customers.

Supports internet connectivity.

It is suited for less confidential information.

Vs



Virtualized resources are shared privately.

There is a cluster of dedicated customers.

Connectivity works over internet, fiber and private network.

It is suited for secured confidential information and core systems.

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Securing the gateways

Network nodes connect two networks with different transmission protocols

- Connection implies vulnerability.
- Can air gap with reduced function.
- Employs one-way diodes.

Zero-trust gateways

- Never trust; always verify.
- Identifies specified control data and protect surface.
- Establish a micro-perimeter with segmentation gateway.
- Understand flows across infrastructure.

Goals

- Improve observability without losing function.
- Implement traffic control.
- Achieve service/resource isolation.
- Maximize SSL/HTTPS connections.



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Automation and scaling

- **Automation:** removes toil and rework.
- **Benefits:** tasks that are self-contained, well-documented, and manually tedious.
- **Provisioning:** reduces time for new environments.
- **Cost reduction:** highlights cost management for a wide range of functions.
- **Capacity planning:** eliminates inconsistencies to reduce complexity.
- **VM sprawl:** keeps active workloads around while slashing non-contributing.

Infrastructure automation

The process of reducing human interaction with IT systems by creating scripts or functions that are repeatable and can be used either by other software or on command (Vmware Glossary).

Infrastructure provisioning only

- Terraform
- Pulumi.

Configuration management

- Ansible
- Chef
- Puppet
- Salt
- Jenkins
- Github
- Docker.



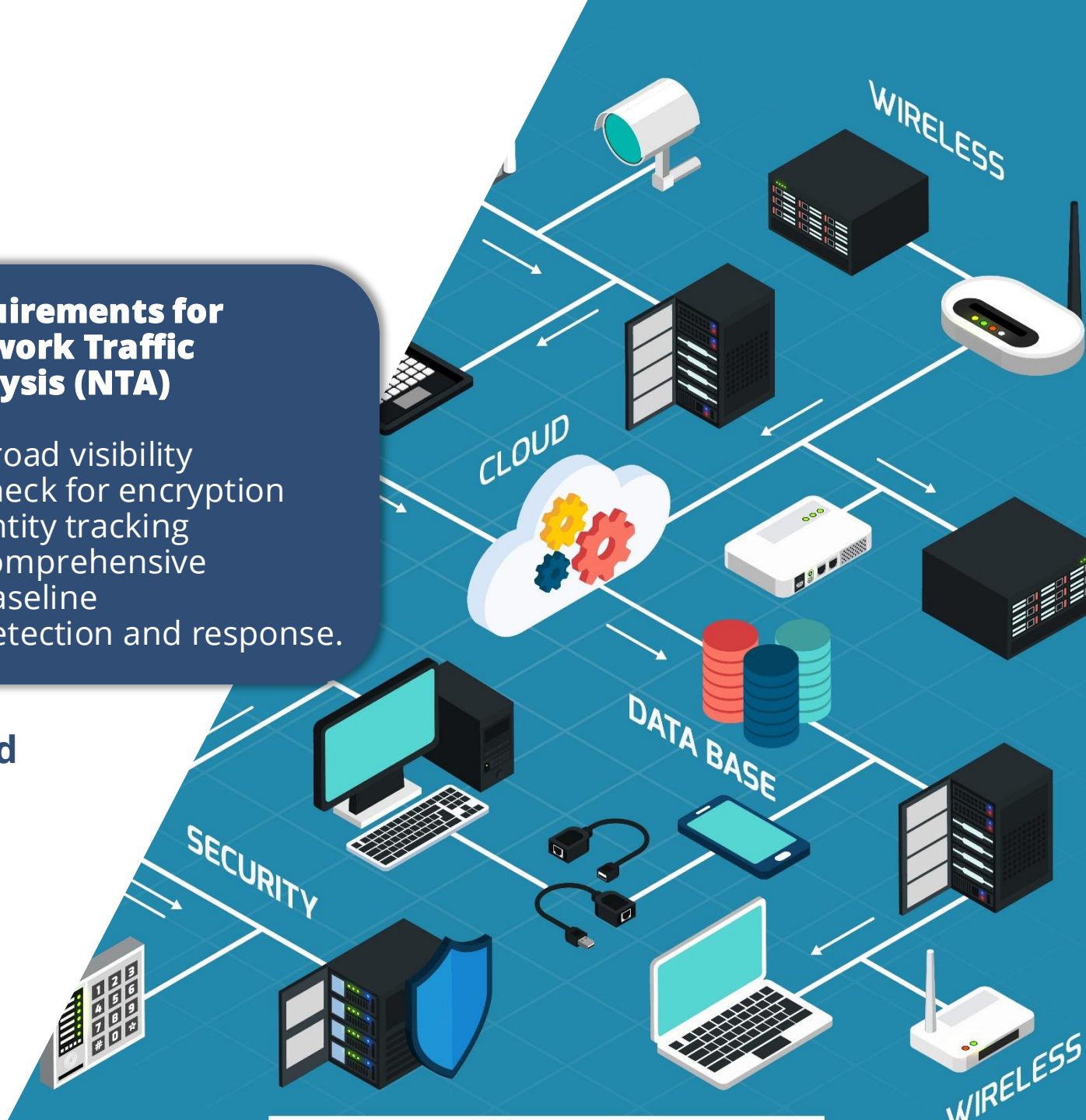
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Traffic analysis

- **Visibility through Open System Interconnection Layers 2-7**
  - north-south traffic
  - east-west traffic
  - TAP/SPAN.
- **Identifies user and organization communication habits**
  - number of terminals per day
  - average call duration.
- **Establish baselines, the known good, and set controls**
  - looks for Indicators of Compromise (IoC)
  - detects faults and maintenance issues.
- **Conversion to MITRE ATT&CK Matrix**
- **Understand remediation options.**

Requirements for Network Traffic Analysis (NTA)

- broad visibility
- check for encryption
- entity tracking
- comprehensive baseline
- detection and response.



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